

Question:

If the sum of all the angles except one of a convex polygon is 2190° , then the number of sides of the polygon must be

- (A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Reasoning + Answer:

<think>

1. Understand the problem:

We are given that the sum of all the angles except one angle of a convex polygon is 2190° .

We need to find the number of sides (n) of the polygon.

2. Use the formula for the sum of interior angles of a polygon:

The sum of the interior angles of a polygon with (n) sides is given by :

$$\text{Sum of interior angles} = (n - 2) \times 180^\circ$$

3. Express the given condition:

Let the missing angle be (x) . According to the problem, the sum of the remaining angles is 2190° . Thus, we have :

$$\text{Sum of all angles} - x = 2190^\circ$$

4. Substitute the sum of the interior angles:

The sum of all the interior angles is :

$$(n - 2) \times 180^\circ$$

So, we can write :

$$(n - 2) \times 180^\circ - x = 2190^\circ$$

5. Solve for (x) :

Since (x) is one of the interior angles, we substitute (x) back into the equation. But first, isolate (x) :

$$x = (n - 2) \times 180^\circ - 2190^\circ$$

6. Ensure (x) is a valid interior angle :

Recall that all interior angles of a convex polygon must be less than 180° .

Therefore, $(x < 180^\circ)$.

This simplifies to :

$$(n - 2) \times 180^\circ - 2190^\circ < 180^\circ$$

7. Solve for (n) :

Add 2190° to both sides :

$$(n - 2) \times 180^\circ < 2190^\circ + 180^\circ$$

$$(n - 2) \times 180^\circ < 2370^\circ$$

Divide both sides by 180° :

$$n - 2 < \frac{2370^\circ}{180^\circ} = 13.1667$$

$$n < 15.1667$$

8. Conclusion :

Since (n) must be an integer, the polygon can have at most 15 sides. Hence, the number of sides (n) is 15.

</think>

<answer> 15 </answer>